Ch.6 Analyzing the Current DB

[Step 2: Analyze current database]

- To understand current ways + deficiencies + future expansions
- Review current data + more interviews with users and managers

➔ Will help you identify preliminary tables and fields

Q1) What types of data does the organization use?

Q2) How does the organization use that data?

Q3) How does the organization manage and maintain that data?

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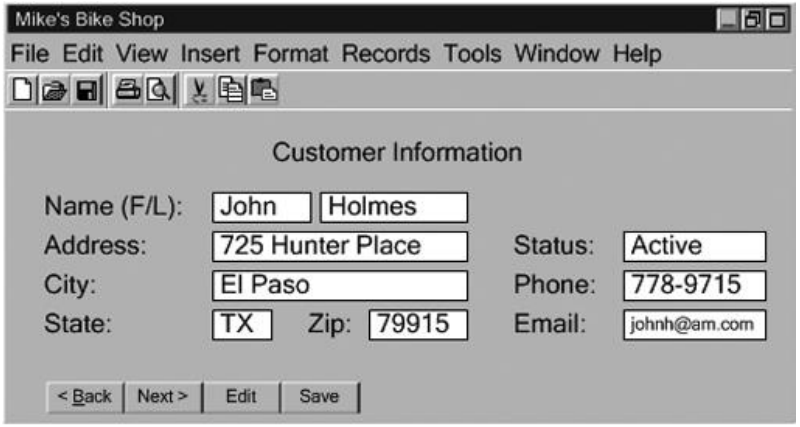
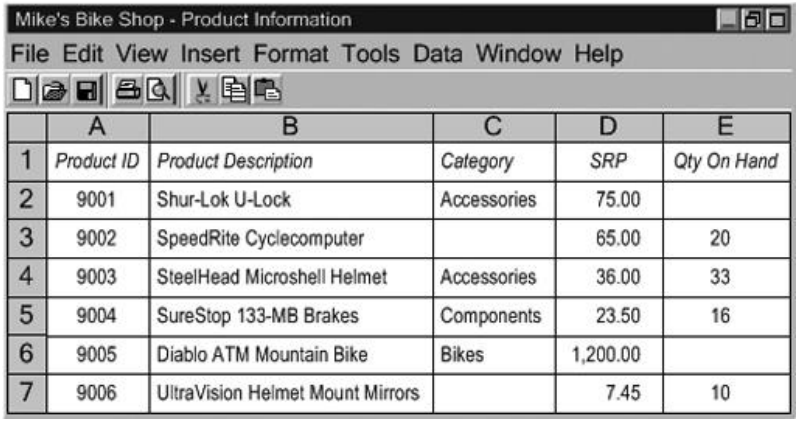
Three main information sources

1. Paper/folders
2. Legacy/Existing database
   - Inherited, or older than 5 years
   - Excel; Word docs
   - Ooops there is a database, adopt it!
   - NO! If this is bug free, they would not hire you to design a new database!
   - Has inefficient structure
3. Human knowledge





	A	B	C	D	E
1	*Product ID*	*Product Description*	*Category*	*SRP*	*Qty On Hand*
2	9001	Shur-Lok U-Lock	Accessories	75.00	
3	9002	SpeedRite Cyclecomputer		65.00	20
4	9003	SteelHead Microshell Helmet	Accessories	36.00	33
5	9004	SureStop 133-MB Brakes	Components	23.50	16
6	9005	Diablo ATM Mountain Bike	Bikes	1,200.00	
7	9006	UltraVision Helmet Mount Mirrors		7.45	10

Figure 6.2 *A typical database screen and a typical spreadsheet screen*

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Analysis 1: Document how is data collected

A1 Office Supplies	
Suite 133	
7739 Alpine Way SE	
Seattle, WA 98115	
Susan Black	519-5883
Email	susanb@A1OS.com

Employee Fact Sheet

Name: George Chavez	Date Hired: June 30, 2009		
Address: 7527 Taxco Drive	City: Seattle	State: WA	Zip: 98115
Phone: 553-0399	Date of Birth: 09/22/85	SSN: 987-65-0049	

Education:

Name of Academic Organization	*Location*	*Year Graduated*
University of Texas at El Paso	El Paso, TX	2007

Figure 6.1 *Examples of paper-based items used to collect data*

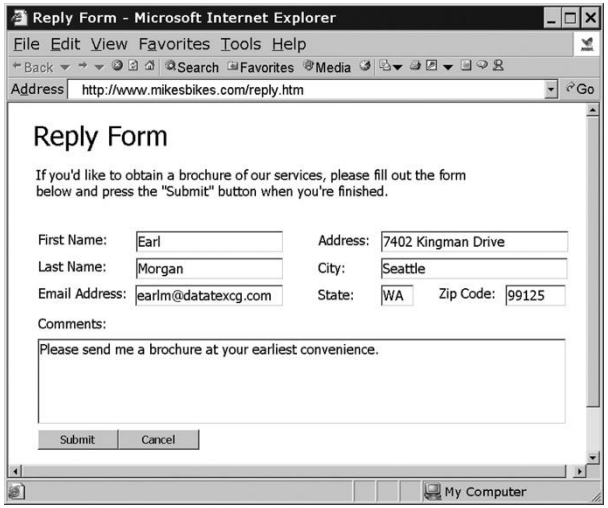

Figure 6.3 *An example of a typical web-based data entry screen*

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Analysis 2: Document how is data presented

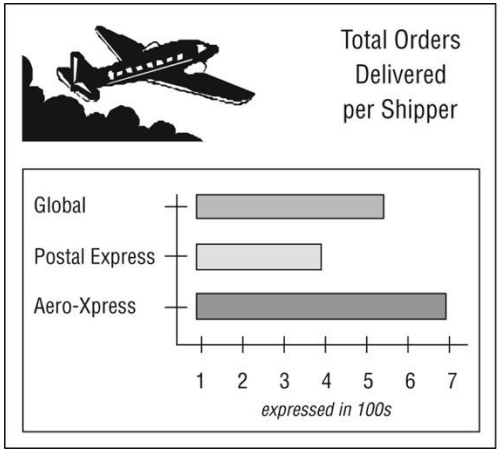
Employee Phone List
as of 05/16/11

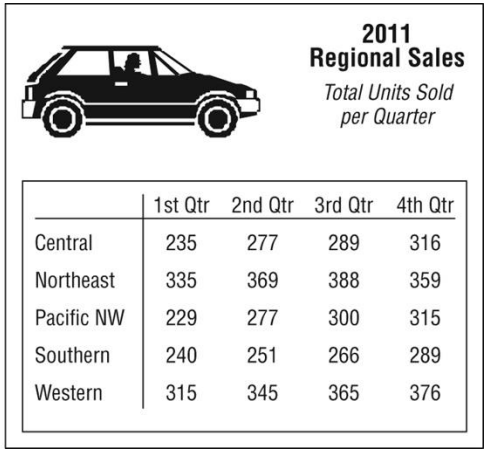
John Alcot	*554-3002*
Regina Allen	*752-5593*
George Chavez	*623-3292*
Ryan Boyd	*554-2991*

Current Product Inventory

Product ID	Product Description	Category	SRP	Quantity
9001	Shur-Lok U-Lock	Accessories	75.00	
9002	SpeedRite Cyclecomputer		65.00	20
9003	SteelHead Microshell Helmet	Accessories	36.00	33
9004	SureStop 133-MB Brakes	Components	23.50	16

Figure 6.4 *A handwritten report and a computer-generated report*





	1st Qtr	2nd Qtr	3rd Qtr	4th Qtr
Central	235	277	289	316
Northeast	335	369	388	359
Pacific NW	229	277	300	315
Southern	240	251	266	289
Western	315	345	365	376

Figure 6.5 *Examples of screen presentation slides*

Figure 6.6 *Example of a web page that presents information from a database*

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Analysis 3: Document how is data managed/used?

- Interview users and managers (asking the good questions is a very important skill)
  1. Open-ended questions ➔ general and reveals any information
  2. Closed-ended questions ➔ to get more specific details

- Subject Identification Technique: look for subjects (nouns) & create a list

> "As an account representative, I'm responsible for ten clients. Each of my clients makes an appointment to come into the showroom to view the merchandise we have to offer for the current season. Part of my job is to answer any questions they have about our merchandise and make recommendations regarding the most popular items. Once they make a decision on the merchandise they'd like to purchase, I write up a sales order for the client. Then I give the sales order to my assistant, who promptly fills the order and sends it to the client."

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Subject-Identification Technique (SIT)

- Look for subjects (nouns) & create an alphabetical list

  "As an account representative, I'm responsible for ten clients. Each of my clients makes an appointment to come into the showroom to view the merchandise we have to offer for the current season. Part of my job is to answer any questions they have about our merchandise and make recommendations regarding the most popular items. Once they make a decision on the merchandise they'd like to purchase, I write up a sales order for the client. Then I give the sales order to my assistant, who promptly fills the order and sends it to the client."

Subjects:

account representative	**merchandise**
appointment	**job**
assistant	**sales order**
clients	**season**
Items	**showroom**

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Characteristic-Identification Technique (CIT)

- Pick a subject and ask specific questions about it
- Look for nouns that are characteristics of this subject

Q: "Let's discuss **sales orders** for a moment. What does it take to complete a sales order for a client?"

A: "Well, I enter all the client information first, such as the client's **name**, **address**, **phone number**, and **email address**. Then I enter the items the client wants to purchase. After I've entered all the items, I tally up the **totals** and I'm done. Oh, I forgot to mention that I enter the client's **fax number** and **shipping address**—if they have one."

Characteristics for **sales orders** (on a separate paper)

address	shipping address
email address	totals
fax number	
name	
phone number	

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Part U) Interview Users

- Interview users and managers (asking the good questions is a very important skill)
- (I1) Open-ended questions ➔ SIT finds **subjects** ➔

Closed-ended questions ➔ CIT finds **characteristics** for each subject ➔

- (I2) Ask specifics about your documented info (SIT-CIT)
- (I3) Ask specifics about your documented report samples (SIT-CIT)
- (I4) Ask specifics about additional info they would like (SIT-CIT)
- (I5) Ask specifics about future new info they would like (SIT-CIT)

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(I1) Questions and SIT&CIT

Case Study:

- INTERVIEWER: "What kind of work do you do on a day-to-day basis?"
- PARTICIPANT: "I accept land-use applications that are submitted by various people, log them in, and set a hearing date with the hearing examiner. I also assist applicants if they have any questions regarding a specific application."
- INTERVIEWER: "Let's talk about the applications for a moment. What types of facts are associated with an application?"
- PARTICIPANT: "There's quite a number, actually. There are facts concerning the type and name of the application, its designation and address, and its location."
- INTERVIEWER: "Tell me about the facts concerning the application's type and name."
- PARTICIPANT: "There are four things we record: the type of application, the name of the subdivision, the purpose of the project, and a description of the project."

Subjects:	Characteristics for applications:
Applicants Applications People	address, description, designation, location, name, purpose, type

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(I2) Interviews about documents (SIT&CIT)

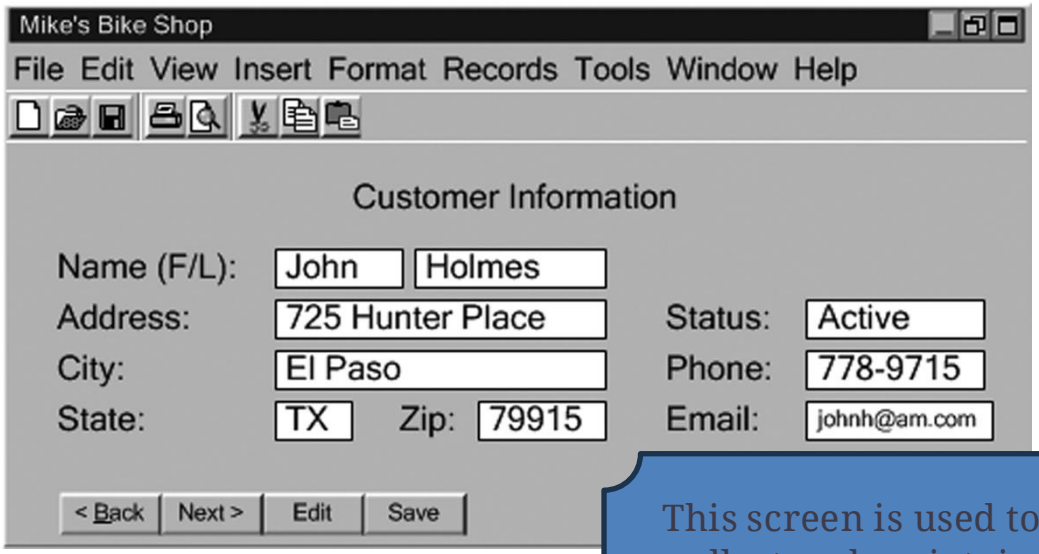
- Next, discuss with users the data you have documented and ask specifics

Q: "You mentioned in a previous discussion that you keep track of all the customers. How does this screen help you to carry out that task?"

A: "This screen allows me to enter new customers, as well as modify and maintain all the information we have on existing customers."

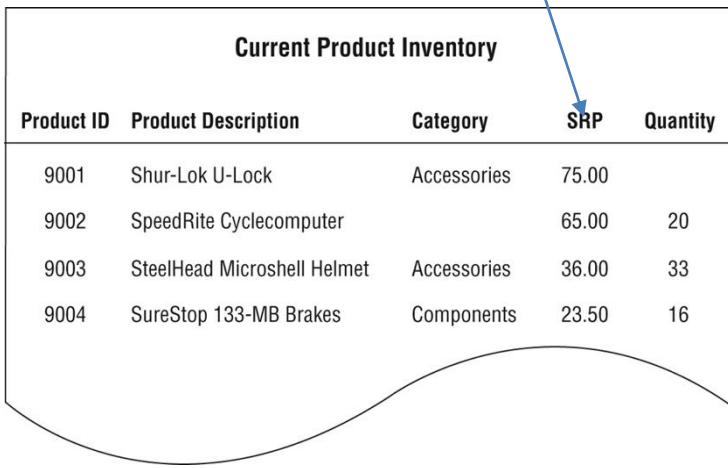
Write succinct label.

Q: What SRP means?



This screen is used to collect and maintain all customer data.

Figure 6.7 *A data collection sample*



Current Product Inventory

Product ID	Product Description	Category	SRP	Quantity
9001	Shur-Lok U-Lock	Accessories	75.00	
9002	SpeedRite Cyclecomputer		65.00	20
9003	SteelHead Microshell Helmet	Accessories	36.00	33
9004	SureStop 133-MB Brakes	Components	23.50	16

Figure 6.8 *A report sample*

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How to label a complex sample?

Q: "What nursing services do you provide besides those listed in this sample?"
A: "We provide various specialized services for the more complex patient. You see only the general services on this report. However, I can show you a complete list of our services that Katherine maintains on her computer."

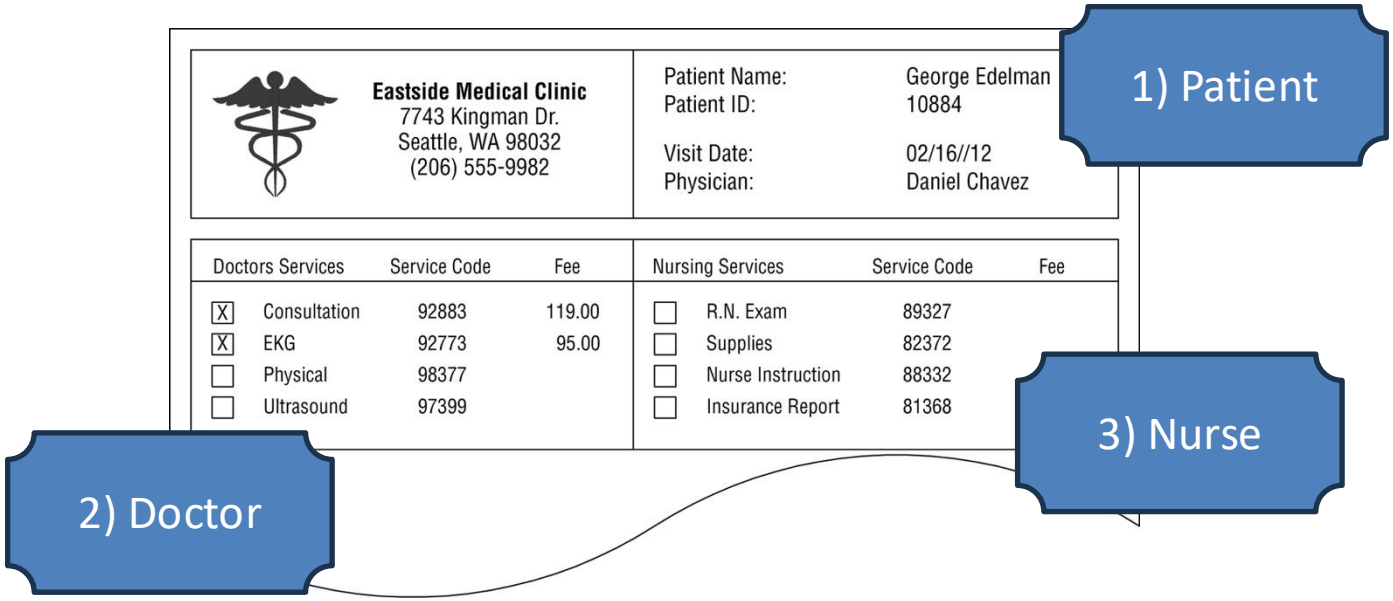

Figure 6.9 *An example of a complex report sample*

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(I3) Interviews about reports (SIT&CIT)

Next, discuss with users report samples

Q: "Do you provide the data that's used to generate this report?"

Q: "Do you personally enter and maintain this data on a daily basis?"

KIRA: "Can you tell me, then, if there is any data that you contribute to the report at all?"

JOAN: "I do supply the customer's name and phone number."

KIRA: "Then you don't supply the customer type or the last purchase date. Is that correct?"

JOAN: "Yes."

KIRA: "Can you tell me who provides this data?" [Follow-up questions are very important!]

JOAN: "I'm not really sure, but . . ."

KIRA: "Do you have an idea of where these items come from?"

JOAN: "As a matter of fact, I do. They come from the sales department."

KIRA: "That sounds good to me. I'll make a note of that on this sample, and then we can move on to the next one."

Customer Phone List

Customer Name	Customer Type	Last Purchase	Phone Number
Alastair Black	Preferred	11/21/11	551-0993
Dave Cunningham	Silver	12/19/11	533-9182
Zachary Ehrlich	Preferred	11/16/11	515-3921
Bill Champlin	Gold	01/22/12	552-3884

Note: Ask Sales about Customer Type

Figure 6.10 *A sample report*

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(I4) Interviews about additional info (SIT&CIT)

Next, ask what additional info the user would like to have on a report

A: "Can we include the vendor's name? It would make it easier to identify a specific product."

- Use SIT(to add subjects to list) → CIT(to add characteristics to list)
- If unclear, settle issues now with follow-up questions

Current Product Inventory

Product ID	Product Description	Category	SRP	Quantity
9001	Shur-Lok U-Lock	Accessories	75.00	
9002	SpeedRite Cyclecomputer		65.00	20
9003	SteelHead Microshell Helmet	Accessories	36.00	33
9004	SureStop 133-MB Brakes	Components	23.50	16

Figure 6.11 *The sample report being reviewed by a participant*

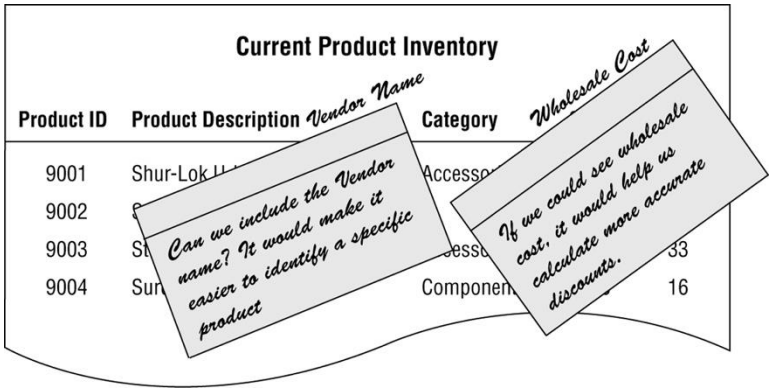

Figure 6.12 *A report sample with a participant's comments*

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(I5) Interviews about future info (SIT&CIT)

Next, ask what future info the user would like to have on a report

Q: How will the organization's evolution affect the amount of information you'll need to do your job?

Q: Do you think you'll need additional types of information to carry out your duties effectively as the organization evolves?

Q: How will the evolution of the organization increase the time you spend on your daily tasks?

Q: Can you predict what types (categories, not specific items) of new information you'll need in order to carry out your duties as the organization evolves?

Q: Do you anticipate a need for new information if your duties are increased as a result of the organization's evolution?

- Together, create such future new report and document it [keep in mind, the user speculates]
- Use SIT(to add subjects to list)

→ CIT(to add characteristics to list)

1st Quarter Customer Sales Statistics

Customer ID	Customer Name	Sales Amounts Maximum	Minimum	Average
9001	Stewart Jameson	265.00	23.00	55.00
9002	Shannon Black	550.00	125.00	70.00
9003	Estela Rosales	250.00	35.00	36.00
9004	Timothy Ennis	325.00	20.00	25.00

Figure 6.13 *An example of a design for a new report*

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Part M) Interview Managers

- (I1) Open-ended questions ➔ SIT finds **subjects** ➔

Closed-ended questions ➔ CIT finds **characteristics** for each subject ➔

- (I2) Ask specifics about your documented info (SIT-CIT)
- (I3) Ask specifics about your documented report samples (SIT-CIT)
- (I4) Ask specifics about additional info they would like (SIT-CIT)
- (I5) Ask specifics about future new info they would like (SIT-CIT)
- [uffff same technique all over again!!!]

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Compile a complete preliminary field list (combines subjects and characteristics)

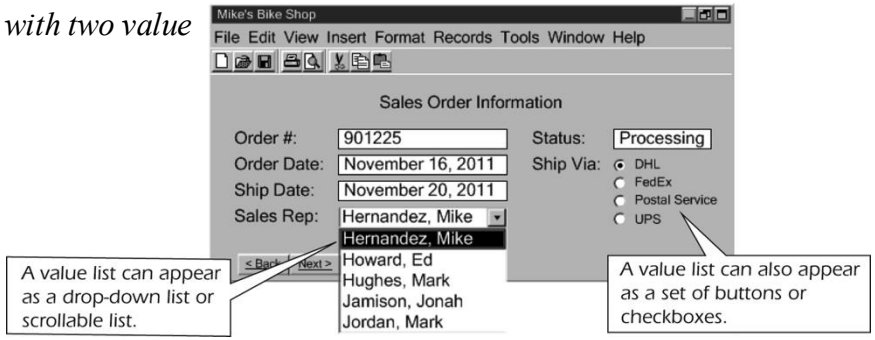
1) Refine
   - –Remove duplicates, or if having different names for the same thing
     - Ex. Product# vs Product No. vs **Product Number**
   - –Refine characteristic names to reflect relationship to a particular subject
     - Ex. **Client** Name; **Employee** Name; **Contact** Name
   - –Check if a characteristic is a subject – ex. "item" is a subject
2) Examine all your documents (again!) to see if you missed any characteristics
   - –for some characteristics, it is good to make a separate list of acceptable values ➔ good for later to create (1) field specification and (2) business rules – see Fig. 6.17 and 6.16 (two possibilities: buttons; drop down list)
3) Create separate list with calculated fields – see ex in Fig. 6.23

Characteristics Incorporating a Value List

Characteristic	Value List
Category	Accessories, Bikes, Clothing, Components, Maintenance, Racks, Wheels
Department	Accessories, Bikes, Clothing, Service,
Sales Rep	The name of every employee within the organization whose position is that of sales rep
Ship Via	DHL, FedEx, Postal Service, UPS

Figure 6.17 *Recording characteristics that incorporate value lists*

Figure 6.16 *A database screen with two value lists*



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Review with owner final preliminary lists of:

(1) fields

(2) calculated fields

(3) [Subjects]

[THIS IS THE OUTPUT OF STEP 2!]

Ex. Mike's Bikes

- Gather information
- Interview users ➔ two lists, subjects and characteristics
- Interview managers ➔ add to the two lists, subjects and characteristics, Fig. 6.22
- Compile **preliminary field list** and **calculated field list**, Fig. 6.23

List of Subjects
as of 02/13/12

Customers	Sales
Employees	Suppliers
Products	

List of Characteristics
as of 02/13/12

Address	Home Phone
Birth Date	Last Name
Category	Name
City	Phone
Comments	Product No.
First Name	State

Figure 6.22 *Partial lists of subjects and characteristics for Mike's Bikes*

Preliminary Field List
as of 02/16/12

Birth Date	Office Phone
Employee Name	Product Name
Employee Address	Category
Employee City	Unit Price
Customer Name	Invoice Number
Customer Address	Invoice Date

Calculated Field List
as of 02/16/12

Discount Amount
Grand Total
Item Total
Subtotal

Figure 6.23 *A partial Preliminary Field List and a Calculated Field List*

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Partial information used on Mike's Bikes database analysis

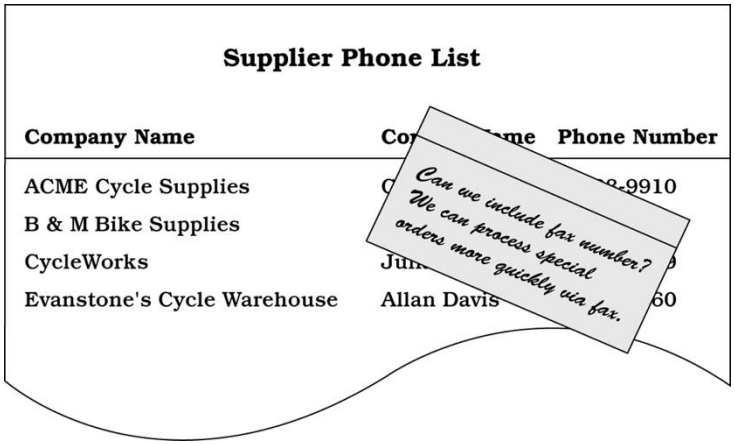
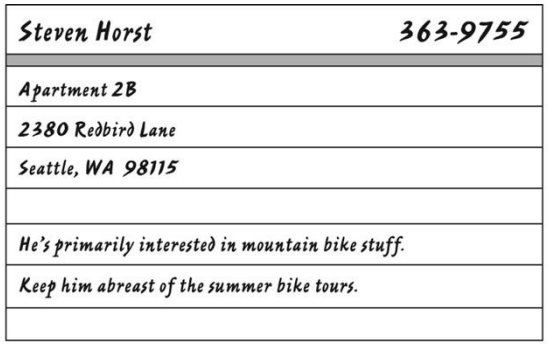
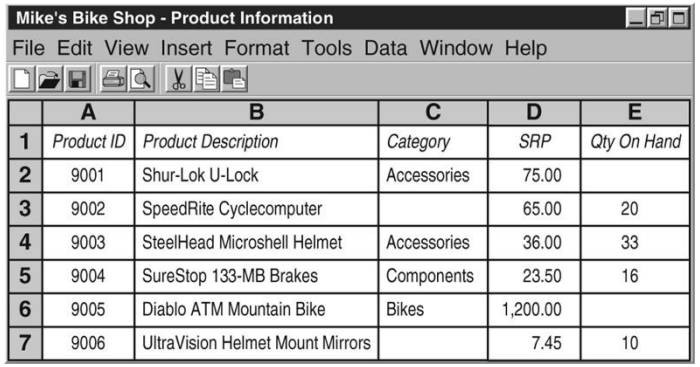


Figure 6.20 *A report sample with attached note suggesting a new field*





	A	B	C	D	E
1	Product ID	Product Description	Category	SRP	Qty On Hand
2	9001	Shur-Lok U-Lock	Accessories	75.00	
3	9002	SpeedRite Cyclecomputer		65.00	20
4	9003	SteelHead Microshell Helmet	Accessories	36.00	33
5	9004	SureStop 133-MB Brakes	Components	23.50	16
6	9005	Diablo ATM Mountain Bike	Bikes	1,200.00	
7	9006	UltraVision Helmet Mount Mirrors		7.45	10

Figure 6.18 *A paper-based and a computer-generated sample from Mike's Bikes*

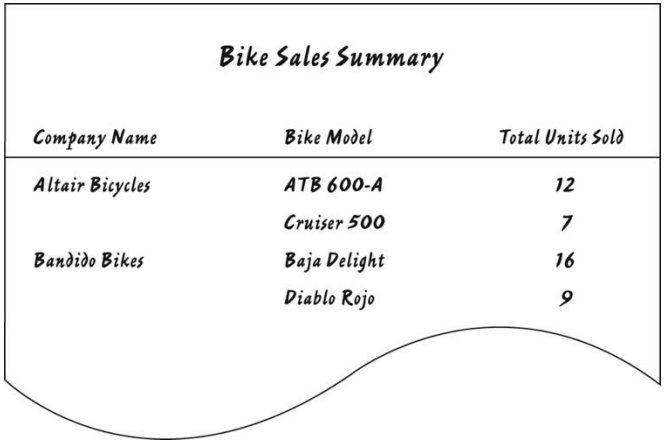


Figure 6.21 *The sketch of Mike's new report*

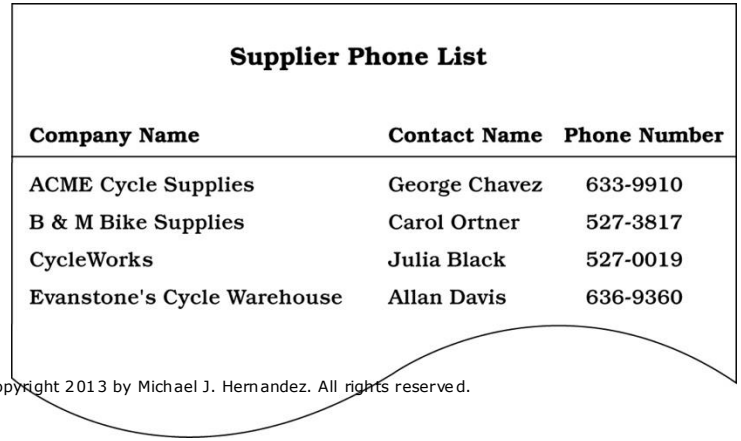


Figure 6.19 *A report sample from Mike's Bikes*

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